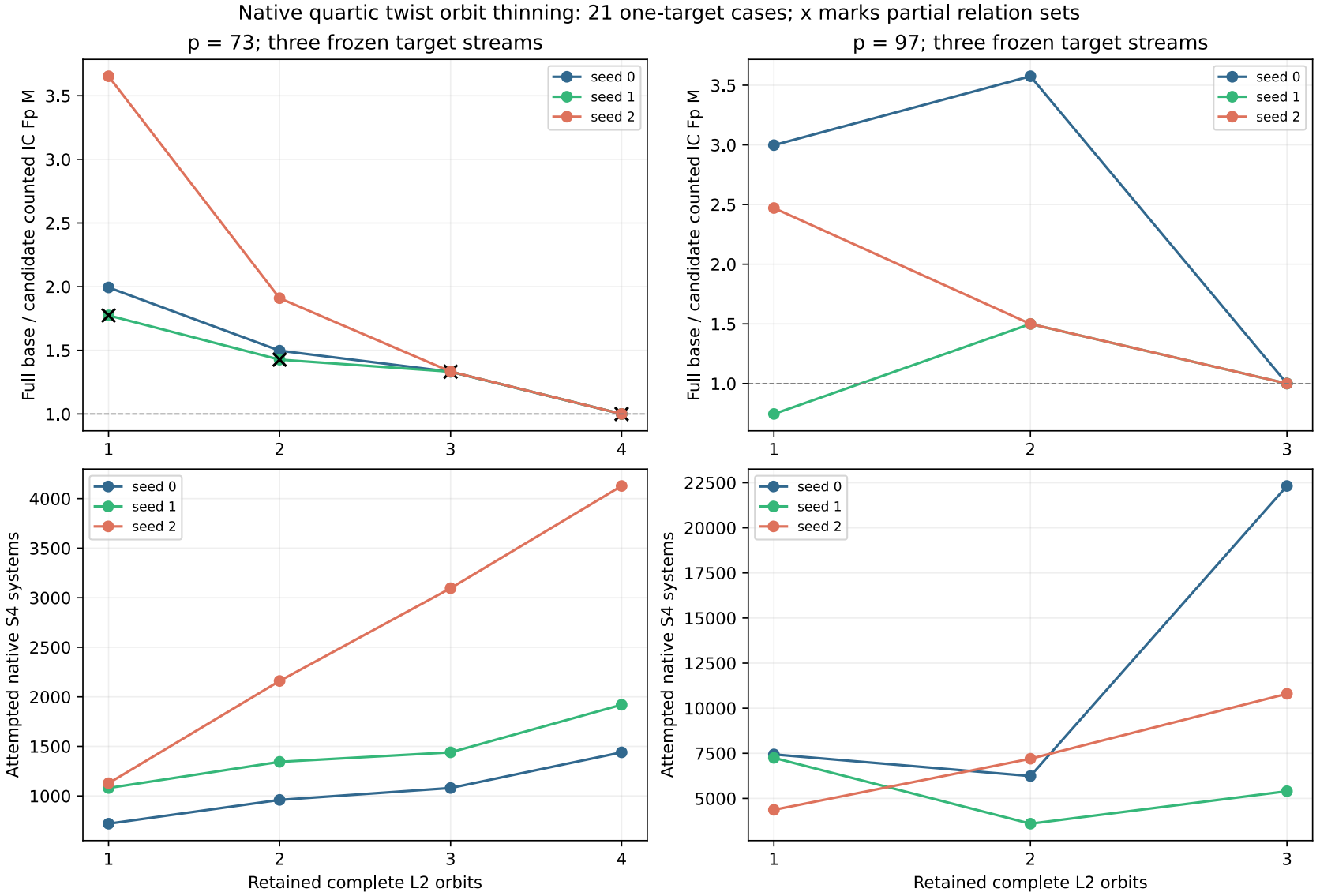
Native orbit-complete quartic-twist polynomial IC

23 October 2026. Public synthetic one-target p=73/97 panel; 21 preregistered native subset cases, three seeds per curve; Q=[271]G. The full Markdown report, frozen protocol, raw records, separate audits and run PDFs are in `K1_THIN_NATIVE_REPORT.md` and `K1_THIN_NATIVE_RUNS.md`.

Mechanism and comparison

One complete L2 twist orbit has 12 physical factors but one folded log column. The k=1 collector peels signed L2 factors and solves S4 on three L0 factors. The full physical base and map are charged even for subsets. Full/candidate below is same-seed counted IC Fp multiplications; IC/rho uses same-target twist24 rho, also in counted Fp multiplications.



Complete native grid

p/seed	j	rank	query	exact/poly	S4 attempts/fail	IC Fp M	full/IC	audit
73/0	1	5	30	6/6	720/0	445,445,605	1.994	pass
73/0	2	6	20	6/6	960/0	593,181,104	1.497	pass
73/0	3	7	15	7/7	1,080/0	666,998,170	1.332	pass
73/0	4	8	15	8/8	1,440/0	888,203,272	1.000	pass
73/1	1	5	45	8/6	1,080/1	667,133,505	1.774	partial
73/1	2	6	28	9/7	1,344/1	829,490,482	1.427	partial
73/1	3	7	20	10/8	1,440/1	888,451,433	1.332	partial
73/1	4	8	20	11/9	1,920/1	1,183,617,228	1.000	partial
73/2	1	5	47	5/5	1,128/0	696,092,141	3.653	pass
73/2	2	6	45	6/6	2,160/0	1,331,570,891	1.910	pass
73/2	3	7	43	11/11	3,096/0	1,907,506,701	1.333	pass
73/2	4	8	43	15/15	4,128/0	2,542,727,892	1.000	pass
97/0	1	4	310	4/4	7,440/0	4,616,280,829	2.998	pass
97/0	2	5	130	8/8	6,240/0	3,870,889,536	3.575	pass
97/0	3	6	310	16/16	22,320/0	13,839,508,536	1.000	pass
97/1	1	4	302	4/4	7,248/0	4,496,535,068	0.745	pass
97/1	2	5	75	5/5	3,600/0	2,233,996,578	1.499	pass
97/1	3	6	75	6/6	5,400/0	3,349,559,070	1.000	pass
97/2	1	4	182	4/4	4,368/0	2,710,417,878	2.471	pass
97/2	2	5	150	5/5	7,200/0	4,465,588,225	1.500	pass
97/2	3	6	150	6/6	10,800/0	6,696,848,851	1.000	pass

All 21 runs reach full anchored rank and independently recover 271. The four p=73 seed-one runs miss two real signed relations from one failed S4 system and remain partial relation finders. First/A/A non-timing records, independent audits, six full-base source parities and a mutation rejection pass.

Claim boundary

Hash/tree and allocator CPU, peak memory, calibrated GAE, isolated ABAB timing and same-input msolve/Magma are missing. Counted field ratios are not priced S or wall-time speed ratios. No matched-rho crossover or deployed-curve claim follows. The complete Markdown table includes separate group counts and IC/rho field-work gaps.